

Crowded Carry in G10 FX and Crypto

Crash risk in FX and crypto carry trades

STUDENTS

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Presentation structure

From the economic idea to two strategies, their risks and a combined portfolio



Reza Zamani

Motivation, questions and hypotheses; Strategy A (G10 FX) and its risk

Paraj Goyal

Data and methodology; Strategy C (crypto) and its risk; combined portfolio

Proposed split · conclusions presented together

Carry pays because it crashes

Same economic mechanism, two very different markets

- High-yield assets earn a premium over low-yield ones.
- The premium is compensation for crash risk: when many investors hold the same trade, forced unwinding produces sudden losses (Brunnermeier, Nagel & Pedersen 2008).
- **Hypothesis: crowding is measurable, and crowded carry should crash more.**

A: G10 FX

C: Crypto

Carry signal	3-month rate differential	Perpetual funding rate
Crowding data	CFTC speculator positions	Funding = leveraged demand
Universe	7 currencies vs USD	12 coins incl. LUNA, FTT
Frequency	Monthly	Weekly rebalance, daily P&L

Research questions and hypotheses

Reza

Stated before running any backtest

Strategy A · G10 FX carry

- RQ1** Does G10 carry earn a premium after costs, and match the academic carry factor?
- RQ2** Does CFTC crowding predict weaker carry, and does cutting exposure help out of sample?
- RQ3** Is carry exposed to volatility shocks and the dollar factor?

Hypotheses

- H1: positive carry premium (H0: zero)
- H1: weaker, more negatively skewed carry after crowded months
- H1: negative beta to volatility innovations

Strategy C · crypto funding carry

- RQ4** Does short-high / long-low funding earn a premium after costs, from funding or price?
- RQ5** What drives its losses, and does a pre-specified risk control help?
- RQ6** Are FX and crypto carry correlated; does combining them diversify?

Hypotheses

- H1: positive premium, mainly from funding (H0: zero)
- H1: little BTC beta; crashes when crowded positions unwind

Data

Paraj

All downloaded 2026-10-02; cleaned snapshot committed in data/clean/

Data	Source	Coverage	Used for
FX spot (7 currencies)	FRED, Fed H.10	1971–2026 (EUR from 1999)	A: returns
3-month interbank rates	OECD via FRED	JPY from 2002, CHF from 1999	A: carry signal
Speculative positioning	CFTC Commitments of Traders	1986–2026 (NZD from 1999)	A: crowding (alt. data)
Prices, volume, funding (12 coins + USDC)	Binance public archive	2020-01 to 2026-08	C: signal and returns
Carry factor portfolios	Lustig–Roussanov–Verdelhan (course file)	1983–2021	Validation, risk factors

1986

CFTC history after matching the pre-2000 'International Monetary Market' contracts

≈ 12–15%

Average annual funding on the large coins (BTC 11.8%, ETH 13.9%, XRP 14.7%)

0.985–1.002

USDC price range: sanity check that the crypto data are read correctly

Methodology

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Research design chosen to keep the tests honest

Pre-specified

All legs, windows, thresholds, lookbacks and costs fixed before the first backtest

In- vs out-of-sample

A: 1999–2010 / 2011–2026. C: 2020–2023 / 2024–2026

Transaction costs

3 bp per unit traded in FX, 5 bp in crypto; gross and net reported

No look-ahead

Weights at t earn $t+1$; CFTC used after its Friday release; funding up to $t-1$

Survivorship

LUNA and FTT kept; a coin trades only with a price and positive volume

Robustness and validation

Grids reported, not used for tuning; A checked against the LRV carry factor

Strategy A: crowding-filtered G10 carry

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JPY, EUR, GBP, CHF, CAD, AUD, NZD vs USD, monthly, 1999–2026

1

Carry portfolio

Each month-end: long the 2 highest-rate currencies, short the 2 lowest. Equal weights, dollar-neutral, 3 bp costs.

2

Crowding signal (alternative data)

CFTC net speculative position / open interest, 36-month z-score, used only after the Friday release.

3

Portfolio crowding

Average z of the long leg minus average z of the short leg.

4

Filter

Crowding > 1: cut positions to half. All parameters fixed before testing; out-of-sample from 2011.

1986

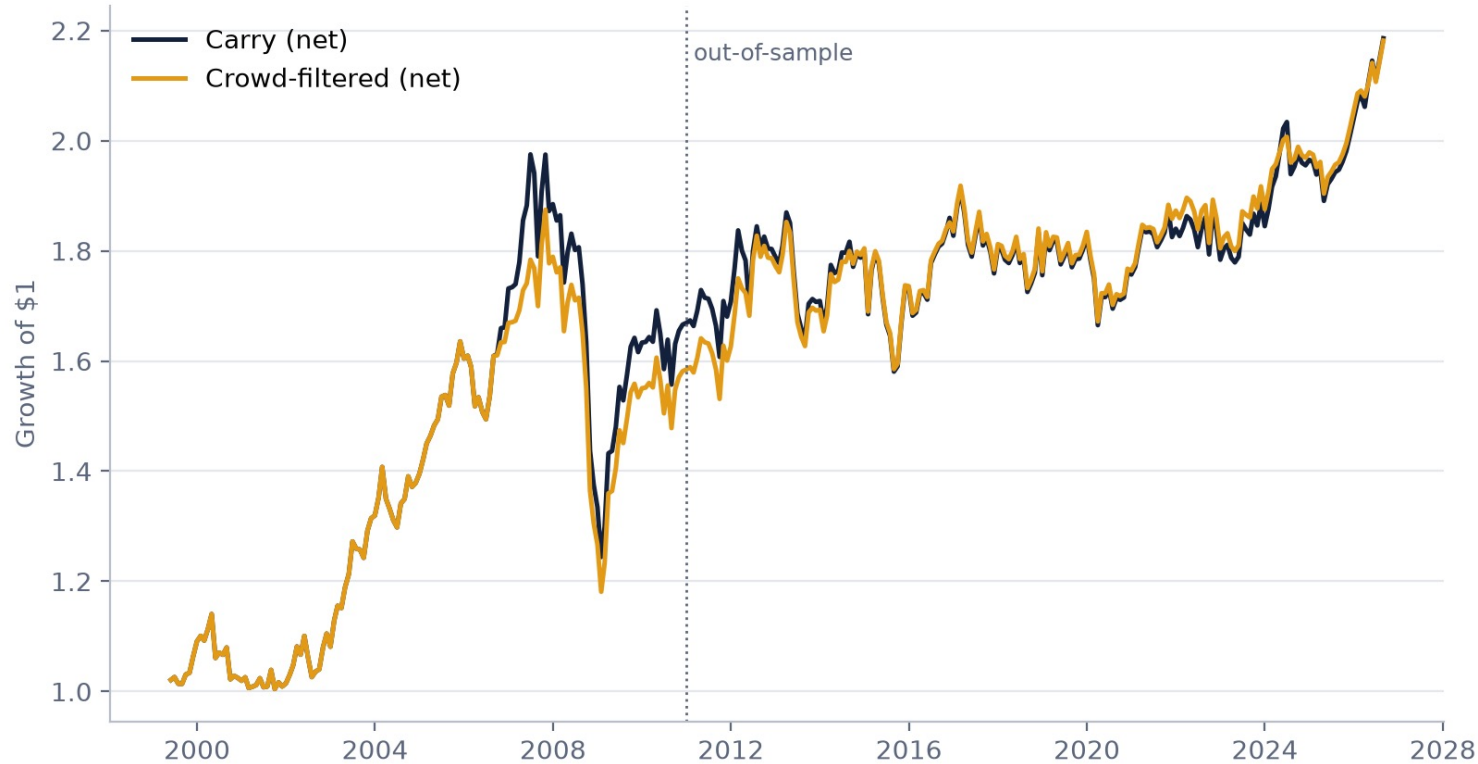
CFTC positioning history (after fixing the pre-2000 CME exchange name)

33

crowded months out of 247 with a crowding signal

Crowding filter helps out of sample, not in 2008

Reza



0.27 → 0.32

Out-of-sample Sharpe, net: carry vs crowd-filtered (2011–2026)

0.30% vs 0.02%

Next-month carry after non-crowded vs crowded months

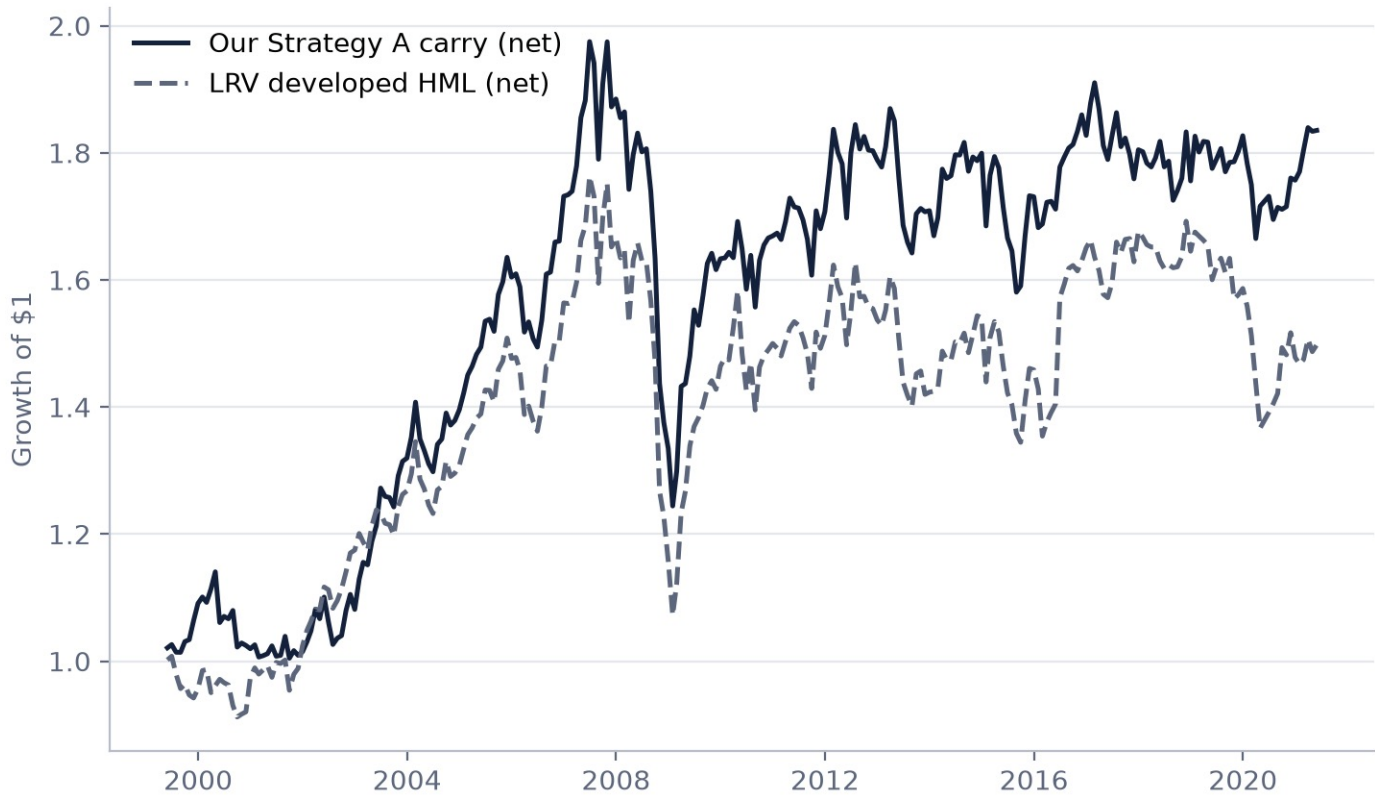
-37%

Max drawdown either way: the 2008 crash began from a non-crowded reading

Our carry matches the academic carry factor

Reza

Strategy A vs Lustig–Roussanov–Verdelhan developed HML, net, 1999–2021



0.82

Monthly correlation with LRV developed HML

Net, 1999–2021	Ours	LRV
Ann. mean (%)	3.2	2.3
Ann. vol (%)	9.5	10.0
Sharpe	0.34	0.23
Max drawdown (%)	-37	-39

Strategy C: crypto funding carry

Paraj

12 Binance USDT perpetuals, 2020–2026; USDC as a sanity check only

1

Signal

Trailing 7-day average funding rate, data up to the day before.

2

Portfolio

Each Sunday: short the top third, long the bottom third by funding. Equal weights, dollar-neutral, 5 bp costs.

3

P&L

Side $\times$ price return – side $\times$ funding paid. A short receives positive funding.

4

No survivorship bias

LUNA and FTT included; a coin trades only on days with a price and volume > 0 .

$\approx 12\text{--}15\%$

Average annual funding on BTC, ETH, XRP, LTC, LINK:
the crypto 'interest rate'

2022-11-14

FTT's last traded day; the archive shows a frozen
price afterwards, which we exclude

Crypto carry: weak, then strong since 2024

Paraj



0.14

Full-sample net Sharpe (4.1% a year, 30% vol)

0.81

Out-of-sample net Sharpe, 2024–2026

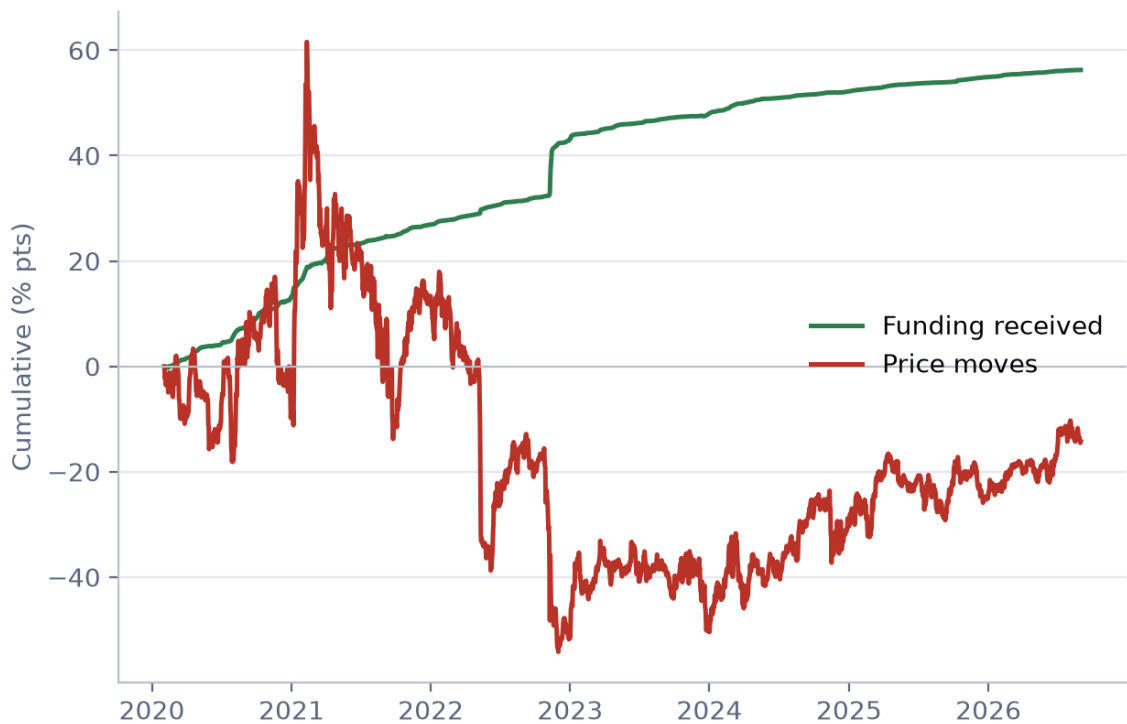
-27%

Worst week (LUNA, May 2022): the strategy was long LUNA

Funding is earned, but price moves dominate

Paraj

Return decomposition and robustness



% per year	Funding	Price	Net
Full	+8.5	-2.2	+4.1
In-sample	+12.3	-12.5	-2.5
Out-of-sample	+3.1	+13.0	+13.9

Lookback × rebalance	Sharpe net	Sharpe OOS
3 days, daily	0.68	0.41
3 days, weekly	0.49	1.29
7 days, daily	0.56	0.92
7 days, weekly (base)	0.14	0.81
30 days, daily	0.28	0.83
30 days, weekly	0.07	0.42

Strategy A loses when volatility spikes

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Monthly net carry regressed on risk factors, Newey-West t-stats

-1.55

Beta to equity-volatility changes (t = -3.7)

0.35

Beta to the dollar factor (t = 4.1)

0.25%

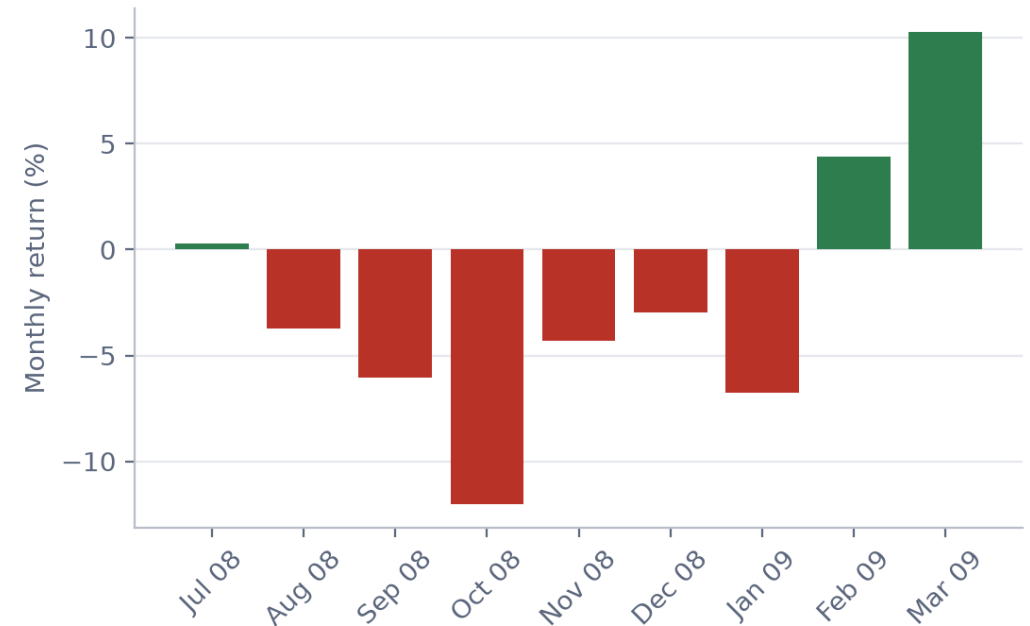
Monthly alpha (t = 1.7): not significant

0.18

R² of the two-factor model

Worst months: Oct 2008, Aug 2007, May 2000, Jan 2009, and the Jan 2015 Swiss franc de-peg.

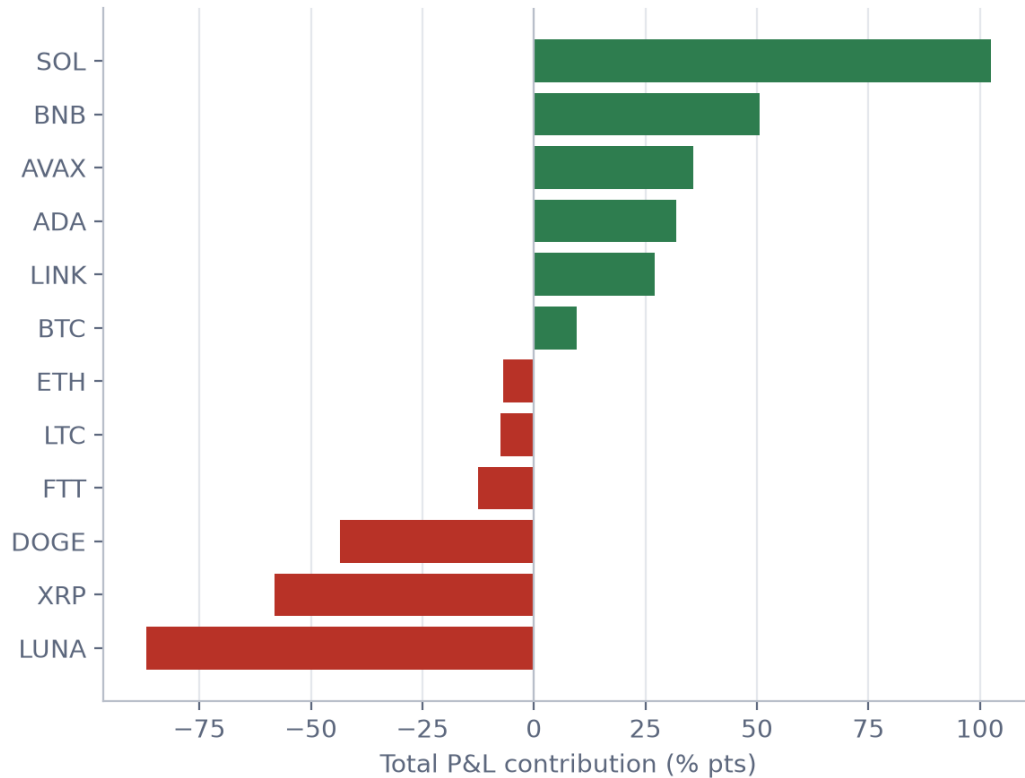
The 2008 carry crash: -20.5% (Jul 2008 – Mar 2009)



A pre-specified risk control did not help

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Cap each coin at 1/6 of a leg; no longs with funding below -50% a year



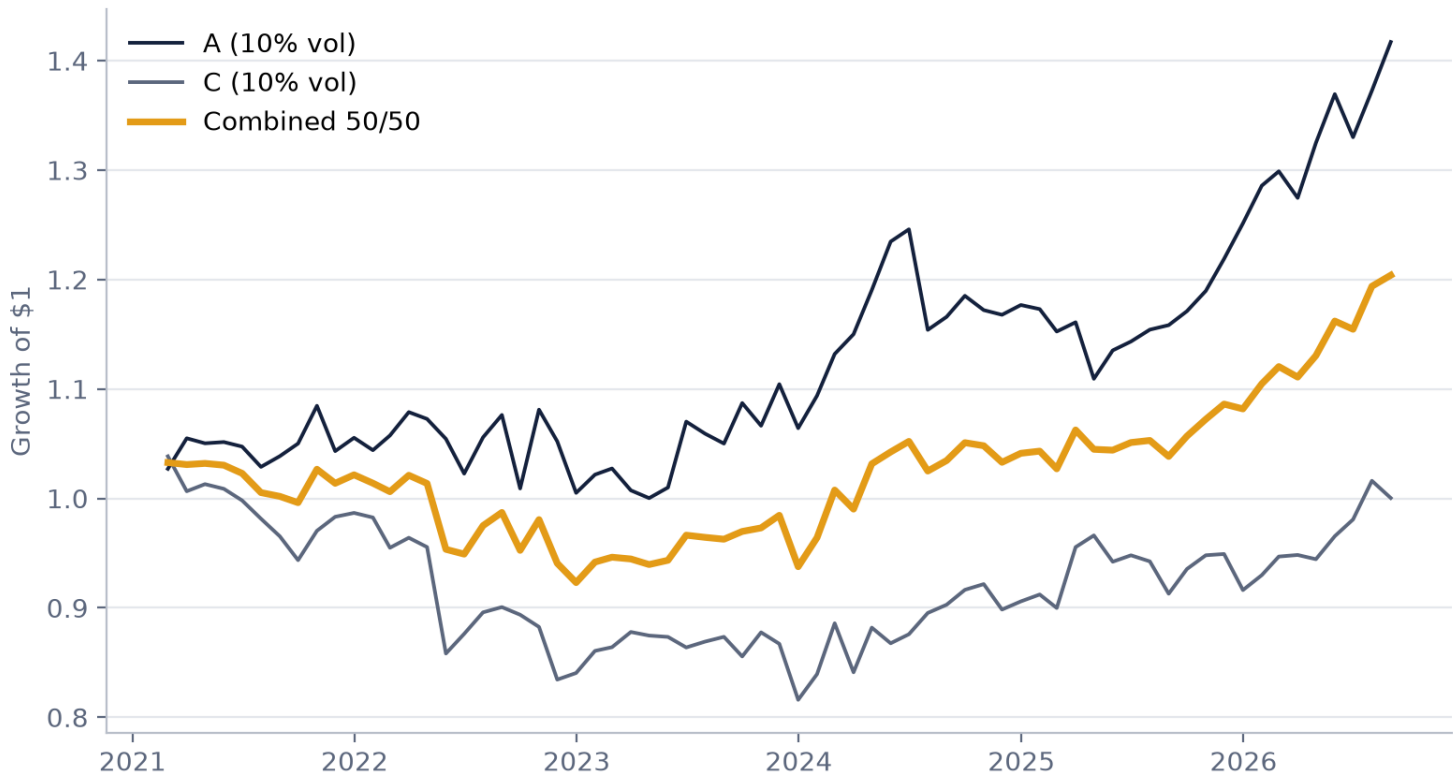
Net	Base	Controlled
Sharpe, full	0.14	-0.16
Sharpe, out-of-sample	0.81	0.74
Max drawdown (%)	-67	-43
LUNA week (%)	-27.3	-18.9

- The cap mostly halves exposure (3–4 coins per leg).
- LUNA's funding crossed -50% mid-week, after the Sunday rebalance.
- The filter blocked the biggest winners, BNB and SOL.
- **Lesson: deeply negative funding means crowded shorts, not distress.**

Combining FX and crypto carry

Paraj

Each scaled to 10% vol with lagged volatility, 50/50, 2021–2026



0.11

Correlation between A and C

	Sharpe	Vol (%)	MaxDD (%)
A	0.71	9.4	-11
C	0.05	9.3	-21
Combined 50/50	0.51	7.0	-11

Conclusions and limitations

Both

What we found

- FX carry reproduces the academic factor (corr 0.82) and is exposed to volatility shocks.
- Crowding predicts weaker carry and modestly improves the out-of-sample Sharpe.
- Crypto funding carry is market-neutral, but its crashes are coin-specific: it ends up long what traders short hardest.
- FX and crypto carry are nearly uncorrelated (0.11).

Limitations

- Short crypto history (2020–2026) and only 12 coins.
- Few crowded FX months (33); the 2008 crash was not flagged.
- Crypto results depend on lookback and rebalancing frequency.
- Costs are simple estimates; no funding-rate caps, borrow limits or exchange risk.